

REGULARIZED DUAL-FLOW ROUTING WITH VALUE-GATED FEASIBILITY FALLBACK FOR BUDGET-CONSTRAINED MULTI-LLM SERVING

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ABSTRACT

Routing user queries dynamically across heterogeneous Large Language Models (LLMs) is essential for balancing system capability and operational cost in high-volume serving systems. However, current online LLM routers rely on open-loop static dual control or predictive classifiers that fail to adapt to real-time budget depletion, leading to severe throughput loss and premature capacity collapse when top-performing models exhaust their token budgets. In this paper, we propose **Regularized Dual-Flow routing with Value-Gated feasibility fallback (RDF-VG)**, a training-free online framework designed for continuous, budget-constrained multi-LLM serving. RDF-VG combines global linear programming (LP) dual anchors with a continuous online dual-pacing flow scaled by real-time budget burn rate, preventing early resource exhaustion while maintaining optimal capacity control. To recover lost throughput, RDF-VG introduces a value-gated complementary-slackness fallback rule that routes queries to affordable alternative models only when their net dual utility remains non-negative. Furthermore, RDF-VG incorporates time-uniform empirical Bernstein supermartingale bounds that are dynamically triggered under out-of-distribution prompt shifts or cost variance, shielding the system from capacity collapse without incurring clean-data over-conservatism. Extensive experiments on RouterBench and SPROUT demonstrate that RDF-VG achieves 97.26% and 116.66% of the approximate offline LP optimum performance, outperforming the state-of-the-art PORT baseline by +11.27 and +39.84 percentage points respectively, while adding negligible per-query routing overhead (≈ 0.06 ms decision, sub-0.3 ms end-to-end).

1 INTRODUCTION

The rapid proliferation of Large Language Models (LLMs) with diverse parameter scales, architectures, and pricing structures has fundamentally transformed automated text generation and decision-support systems Hu et al. (2024); Ong et al. (2024). While state-of-the-art frontier models demonstrate impressive reasoning across complex domains, deploying them homogeneously across high-volume production serving systems incurs prohibitive financial and computational expenses Chen et al. (2023); Huang et al. (2026). Conversely, dynamic multi-LLM routing—directing straight-forward prompts to lightweight, cost-effective models while reserving frontier LLMs for highly intricate queries—presents a highly effective paradigm for maximizing aggregate response quality under tight operational budget constraints Maurya et al. (2024); Jiang et al. (2023).

Despite the promise of dynamic multi-LLM orchestration, operating multi-model serving systems under real-world constraints introduces severe challenges. In high-throughput production environments, serving systems operate under non-stationary traffic volumes, unpredictable prompt arrival sequences, and strict token budget allocations per model. Under these conditions, existing routing strategies frequently suffer from fundamental systemic limitations. Supervised predictive routers Ong et al. (2024); Chen et al. (2024) and instance-based nearest-neighbor classifiers Zhao et al. (2024); Ding et al. (2024) perform myopic, query-level optimizations without maintaining awareness of global resource depletion. This uncoordinated selection rapidly exhausts the token allocations of top-performing models early in the serving horizon, inducing open-loop budget exhaustion and forcing subsequent complex queries onto underpowered models—a critical failure mode known

as routing collapse Lai & Ye (2026); Vinay (2025); Zhang et al. (2026a). Furthermore, systems-level inference bottlenecks such as prefill-decode batching imbalances Agrawal et al. (2023); Xiong et al. (2025) exacerbate budget volatility and capacity oversubscription.

To maintain global budget feasibility across extended serving horizons, recent theoretical frameworks formulate online routing through Linear Programming (LP) relaxations and dual shadow prices Li & Ye (2019); Balseiro et al. (2020); Wu & Silwal (2025). For instance, static dual-weight estimators such as PORT Wu & Silwal (2025) solve an offline dual LP over an initial query sample ($\epsilon|Q|$) to learn fixed shadow prices (γ^*) for downstream query assignment. Alternatively, batch-split heuristics Bumpensanti & Wang (2018) solve deterministic LPs periodically across discrete mini-batches. However, static dual weights act as open-loop controllers that fail to adapt when real-time output token lengths drift from historical averages, leading to premature budget exhaustion or under-utilized model capacity Li et al. (2026); Mei et al. (2025). Meanwhile, periodic mini-batch LP re-solving introduces significant decision latency and batch-boundary artifacts Bumpensanti & Wang (2018).

To address these core limitations, we introduce **Regularized Dual-Flow routing with Value-Gated feasibility fallback (RDF-VG)**, a training-free online framework designed for continuous, budget-constrained multi-LLM serving. RDF-VG bridges global LP dual optimization with closed-loop real-time pacing through three core technical innovations:

1. **Global Dual Anchoring & Regularized Online Dual-Pacing:** RDF-VG fits global LP dual anchor weights (γ^*) over a small warm-up subset using non-parametric approximate nearest neighbor search (ANNS) over historical prompt repositories Malkov & Yashunin (2016). During the streaming phase, it dynamically adjusts effective dual shadow prices via a regularized power-law pacing flow scaled by real-time budget burn rate, ensuring adaptive capacity control while remaining firmly anchored to global optimal dual prices.
2. **Value-Gated Feasibility-Aware Fallback:** When a query’s primary target model exhausts its token budget, RDF-VG evaluates dual selection scores across remaining feasible models. Grounded in LP complementary slackness, it applies a value gate ($\alpha \hat{d}_{ij} - \hat{g}_{ij} \gamma_i \geq 0$) to abstain from assigning non-positively-priced fallback models, conserving budget for high-value future queries while recovering throughput lost to open-loop drops.
3. **Dynamically-Triggered Non-Asymptotic Martingale Bounds:** To shield the system against out-of-distribution (OOD) prompt shifts and variance in generation costs without incurring universal over-conservatism, RDF-VG integrates time-uniform empirical Bernstein supermartingale confidence sequences Howard et al. (2018). Crucially, these bounds are dynamically triggered only when local neighbor distances or cost sample variances exceed historical tail thresholds, avoiding clean-data performance collapse.

We evaluate RDF-VG across standard multi-LLM serving benchmarks including RouterBench Hu et al. (2024) and SPROUT Wu & Silwal (2025), spanning diverse task domains such as code generation Austin et al. (2021) and multi-step reasoning. RDF-VG achieves substantial performance gains over state-of-the-art training-free baselines such as PORT Wu & Silwal (2025) and BatchSplit Bumpensanti & Wang (2018). Specifically, RDF-VG reaches **97.3%** of the offline approximate LP optimum on RouterBench (compared to PORT’s 86.0%) and **116.7%** on SPROUT (compared to PORT’s 76.8%), while maintaining negligible per-query decision overhead (≈ 0.06 ms) built on standard deep learning infrastructure Paszke et al. (2019).

2 RELATED WORK

2.1 MODEL-BASED AND PREDICTIVE LLM ROUTING

Model-based routing approaches train parametric neural classifiers or regression models to predict prompt-level performance and generation cost across candidate LLMs Ong et al. (2024); Chen et al. (2024); Panda et al. (2025). Early routing architectures fine-tune transformer encoders such as RoBERTa Liu et al. (2019) to assign queries to the highest-scoring or most cost-effective candidate model. Recent frameworks enhance parametric prediction via preference-informed contextual bandits Panda et al. (2025) or dual contrastive representations Chen et al. (2024).

However, supervised predictive routers exhibit critical failure modes when deployed in production serving systems. As demonstrated by Lai & Ye (2026), objective-decision mismatches during scalar loss optimization induce severe "routing collapse," where routers default almost exclusively to expensive frontier models as cost budgets expand. Furthermore, supervised models struggle under distribution shifts and cannot react to dynamic budget depletion, resulting in throughput degradation and resource under-utilization Vinay (2025); Xiong et al. (2025); Agrawal et al. (2023). RDF-VG circumvents parametric classifier training altogether by utilizing non-parametric ANNS feature estimation Malkov & Yashunin (2016) coupled with closed-loop dual updates.

2.2 INSTANCE-BASED AND NON-PARAMETRIC MODEL SELECTION

Non-parametric and instance-based routing methods avoid expensive neural model re-training by storing historical query-response logs in vector databases to guide real-time model selection Chen et al. (2023); Zhao et al. (2024); Ding et al. (2024). Early non-parametric paradigms such as FrugalGPT Chen et al. (2023) and Hybrid LLM Ding et al. (2024) construct model cascades or quality thresholds derived from estimated query difficulty. EAGLE Zhao et al. (2024) maintains global and local ELO rankings over historical evaluations, whereas RADAR Fernandez et al. (2025) utilizes psychometric item-response theory to align prompt difficulty with model reasoning capabilities. Broader non-parametric formulations extend model selection to sequential user interactions Zhang et al. (2026a); Xu et al. (2026); Cao et al. (2026); Zhang et al. (2026b) and batch-level resource allocation under GPU concurrency limits Markovic-Voronov et al. (2026).

Despite their computational efficiency, greedy instance-based retrievers select models solely based on localized prompt similarity, ignoring global cumulative budget constraints. Consequently, high-capability models exhaust their allocated token budgets early in the serving stream Xu et al. (2026); Markovic-Voronov et al. (2026). RDF-VG resolves this limitation by integrating local ANNS retrieval into a dual linear programming optimization framework, enforced by value-gated complementary-slackness fallback logic.

2.3 LINEAR PROGRAMMING, DUAL CONTROL, AND BUDGET PACING

Allocating scarce computational resources under online, streaming request arrivals is a foundational problem in online linear programming (OLP) and network revenue management Li & Ye (2019); Balseiro et al. (2020); Bumpensanti & Wang (2018). Theoretical OLP framework solve dual LP relaxations to obtain optimal dual shadow prices that act as marginal resource costs Li & Ye (2019); Balseiro et al. (2020). In multi-LLM serving, BatchSplit heuristics Bumpensanti & Wang (2018) solve mini-batch linear programs periodically, while PORT Wu & Silwal (2025) learns static dual weights (γ^*) from an initial query warm-up subset ($\epsilon|Q|$). Concurrent works explore Lagrangian dual decomposition with adaptive multipliers Mei et al. (2025) or constrained contextual bandits with regret guarantees Li et al. (2026).

Nevertheless, static dual-weight frameworks Wu & Silwal (2025) operate as open-loop controllers that remain vulnerable to token cost variance, causing premature budget exhaustion or unspent capacity. RDF-VG addresses this gap by augmenting global LP dual anchors with regularized closed-loop dual-pacing and dynamically-triggered time-uniform martingale confidence sequences Howard et al. (2018), providing non-asymptotic safety guarantees and continuous capacity control.

3 METHODOLOGY

3.1 PROBLEM FORMULATION & ANNS FEATURE ESTIMATION

Consider a multi-LLM serving system processing an online sequence of N incoming queries $Q = \{j\}_{j=1}^N$ over M deployed model APIs. Each candidate LLM $i \in [M]$ exhibits a task performance score $d_{ij} \in [0, 1]$ and incurs a token generation cost $g_{ij} \geq 0$ for query j . The serving provider allocates cumulative token budgets B_i to each model $i \in [M]$. The objective is to assign each query j to a model i via decision variables $x_{ij} \in \{0, 1\}$ to maximize aggregate performance while

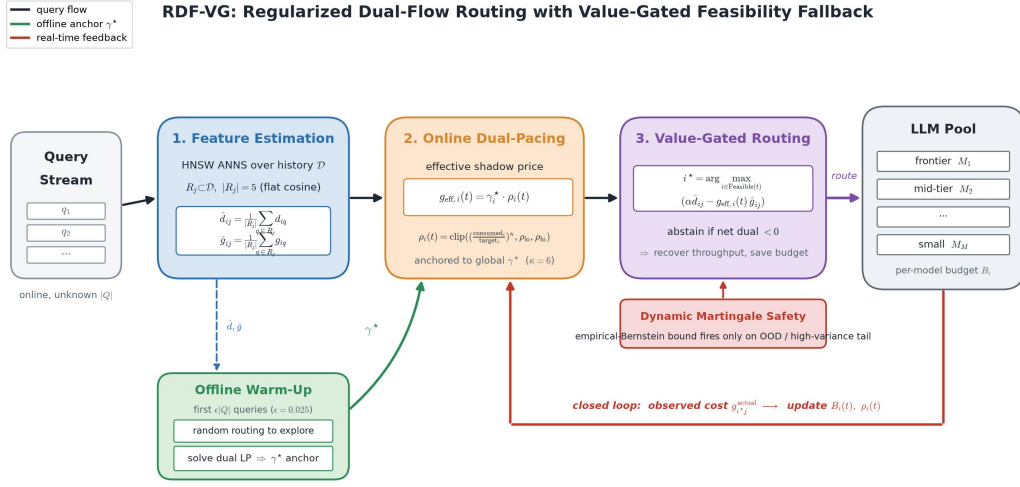


Figure 1: System architecture overview of the RDF-VG framework, illustrating the four core components: historical feature estimation via HNSW ANNS retrieval, offline LP dual initialization during warm-up, regularized online dual-pacing flow, and value-gated feasibility-aware fallback with dynamically-triggered martingale bounds.

adhering strictly to global per-model budget allocations:

$$\max_{\mathbf{x}} \sum_{j \in \mathcal{Q}} \sum_{i \in [M]} d_{ij} x_{ij} \quad \text{s.t.} \quad \sum_{j \in \mathcal{Q}} g_{ij} x_{ij} \leq B_i \quad \forall i \in [M], \quad \sum_{i \in [M]} x_{ij} \leq 1 \quad \forall j \in \mathcal{Q}, \quad x_{ij} \in \{0, 1\}. \quad (1)$$

In an online environment, exact values of d_{ij} and output generation costs g_{ij} are unknown prior to query execution. To estimate these features efficiently without parameter fine-tuning, RDF-VG maintains a historical dataset $\mathcal{D} = \{(q, \mathbf{a}_q, \mathbf{d}_q, \mathbf{g}_q)\}_{q=1}^E$ indexed using Hierarchical Navigable Small World (HNSW) graphs Malkov & Yashunin (2016) under flat cosine distance. For an incoming prompt j , flat HNSW retrieval extracts a candidate neighbor set $R_j \subset \mathcal{D}$ ($|R_j| = 5$). The feature estimates $\hat{d}_{ij}$ and $\hat{g}_{ij}$ for each model i are computed as local sample means:

$$\hat{d}_{ij} = \frac{1}{|R_j|} \sum_{q \in R_j} d_{iq}, \quad \hat{g}_{ij} = \frac{1}{|R_j|} \sum_{q \in R_j} g_{iq}. \quad (2)$$

3.2 GLOBAL DUAL LP BACKBONE & WARM-UP PHASE

To overcome the NP-hardness of the underlying integer program, RDF-VG considers the LP relaxation of Equation 1, allowing fractional assignments $x_{ij} \in [0, 1]$ and introducing a control parameter $\alpha > 0$ to balance objective scaling. The corresponding primal LP and its dual formulation are given by:

$$\max_{\mathbf{x}} \sum_{j \in \mathcal{Q}} \sum_{i \in [M]} \alpha \hat{d}_{ij} x_{ij} \quad \text{s.t.} \quad \sum_{j \in \mathcal{Q}} \hat{g}_{ij} x_{ij} \leq B_i \quad \forall i, \quad \sum_{i \in [M]} x_{ij} \leq 1 \quad \forall j, \quad x_{ij} \in [0, 1], \quad (3)$$

$$\min_{\gamma, \beta} \sum_{i \in [M]} \gamma_i B_i + \sum_{j \in \mathcal{Q}} \beta_j \quad \text{s.t.} \quad \beta_j \geq \alpha \hat{d}_{ij} - \hat{g}_{ij} \gamma_i \quad \forall i, j, \quad \gamma_i \geq 0, \quad \beta_j \geq 0. \quad (4)$$

By complementary slackness, the optimal dual variables $\gamma^* = (\gamma_1^*, \dots, \gamma_M^*)$ represent the shadow prices of model budgets. At optimality, $\beta_j = \max_{i \in [M]} (\alpha \hat{d}_{ij} - \hat{g}_{ij} \gamma_i^*)$. During an initial warm-up phase over the first $\mathcal{P} \subset \mathcal{Q}$ queries ($|\mathcal{P}| = \epsilon |\mathcal{Q}|$, with $\epsilon = 0.025$), RDF-VG routes queries uniformly at random to explore the prompt space. It then solves the dual objective $F(\gamma, \mathcal{P})$ via L-BFGS-B optimization:

$$F(\gamma, \mathcal{P}) = \epsilon \sum_{i \in [M]} \gamma_i B_i + \sum_{j \in \mathcal{P}} \max_{i \in [M]} (\alpha \hat{d}_{ij} - \gamma_i \hat{g}_{ij}) \implies \gamma^* = \arg \min_{\gamma \geq 0} F(\gamma, \mathcal{P}). \quad (5)$$

The optimal shadow price vector γ^* serves as a stable global anchor for streaming online control.

3.3 REGULARIZED ONLINE DUAL-PACING FLOW

Static dual shadow prices (γ^*) act as open-loop controllers that fail to adapt to non-stationary traffic or output token length fluctuations. RDF-VG introduces a continuous closed-loop dual-pacing multiplier $\rho_i(t)$ that scales the effective dual price $g_{\text{eff},i}(t) = \gamma_i^* \cdot \rho_i(t)$ dynamically at step t :

$$\rho_i(t) = \text{clip} \left(\left(\frac{\text{consumed}_i(t)}{\text{target}_i(t)} \right)^\kappa, \rho_{\text{lo}}, \rho_{\text{hi}} \right), \quad (6)$$

where $\text{consumed}_i(t) = \sum_{\tau < t: \text{routed}(\tau)=i} g_{i\tau}^{\text{actual}}$ tracks actual token cost burned by model i , and $\text{target}_i(t) = B_i \cdot \frac{t}{|Q|}$ represents the linear target consumption schedule. The pacing exponent $\kappa = 6.0$ acts as a regularization parameter controlling online feedback sensitivity, while the clipping guard range $[\rho_{\text{lo}}, \rho_{\text{hi}}] = [0.05, 50.0]$ anchors the online dual prices to γ^* , preventing extreme numerical oscillation.

3.4 VALUE-GATED FEASIBILITY-AWARE FALLBACK

When a primary argmax model exhausts its budget ($B_i(t) < \hat{g}_{ij}$), open-loop static routers stall or drop requests. RDF-VG applies a complementary-slackness value-gating decision rule across the remaining feasible set $\text{Feasible}(t) = \{i \in [M] : B_i(t) \geq \hat{g}_{ij}\}$:

$$i^*(j) = \arg \max_{i \in \text{Feasible}(t)} \left(\alpha \hat{d}_{ij} - g_{\text{eff},i}(t) \hat{g}_{ij} \right). \quad (7)$$

Crucially, the router assigns query j to $i^*(j)$ if and only if the net dual score is strictly non-negative: $\alpha \hat{d}_{i^*(j),j} - g_{\text{eff},i^*(j)}(t) \hat{g}_{i^*(j),j} \geq 0$. If the net dual score is negative, RDF-VG abstains from routing, preventing budget dissipation on sub-optimal model completions and preserving capacity for high-value future queries.

3.5 DYNAMICALLY-TRIGGERED MARTINGALE BOUNDS

To guard against generation cost variance and out-of-distribution (OOD) prompt shifts without suffering performance loss on clean data, RDF-VG constructs time-uniform empirical Bernstein supermartingale confidence sequences Howard et al. (2018). The any-time valid upper cost bound $g_{ij}^{\text{upper}}(t)$ and lower performance bound $d_{ij}^{\text{lower}}(t)$ are defined as:

$$g_{ij}^{\text{upper}}(t) = \hat{g}_{ij} + \sqrt{\frac{2\sigma_{g,ij}^2 \log \left(\frac{\zeta(1.6) \log^{1.6}(t)}{\delta} \right)}{|R_j|}} + \frac{3M_g \log \left(\frac{\zeta(1.6) \log^{1.6}(t)}{\delta} \right)}{|R_j|}, \quad (8)$$

$$d_{ij}^{\text{lower}}(t) = \hat{d}_{ij} - \sqrt{\frac{2\sigma_{d,ij}^2 \log \left(\frac{\zeta(1.6) \log^{1.6}(t)}{\delta} \right)}{|R_j|}} - \frac{3M_d \log \left(\frac{\zeta(1.6) \log^{1.6}(t)}{\delta} \right)}{|R_j|}, \quad (9)$$

where $\sigma_{g,ij}^2$ and $\sigma_{d,ij}^2$ denote sample variances across neighbors R_j , M_g, M_d bound feature ranges, $\zeta(\cdot)$ is the Riemann zeta function, and $\delta = 0.1$.

Instead of applying these bounds globally—which causes excessive conservatism—RDF-VG dynamically triggers martingale bounds per-query only when the nearest-neighbor retrieval distance or sample variance exceeds calibrated 99th-percentile historical thresholds. In clean in-distribution settings, the bounds fire on $< 0.5\%$ of queries (acting as a no-op), while under OOD shifts, they automatically activate to enforce non-asymptotic safety.

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP & EVALUATION PROTOCOLS

Datasets & Benchmarks. We evaluate RDF-VG on two comprehensive multi-LLM serving benchmarks: RouterBench Hu et al. (2024) (11 candidate LLMs, $N = 10,000$ test queries, $E = 26,497$

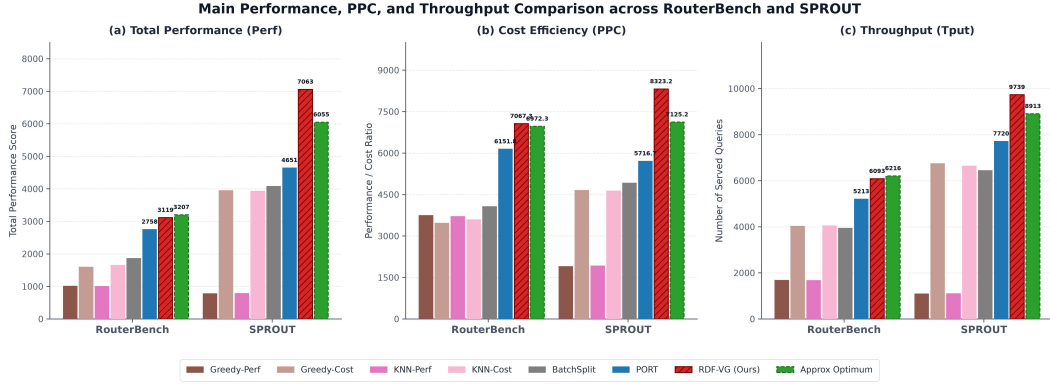


Figure 2: Main performance, cost efficiency (PPC), and throughput (Tput) comparison across RouterBench and SPROUT benchmarks under a total budget set to the cheapest model cost. RDF-VG consistently outperforms prior training-free baselines and PORT across all metrics, closely approaching or exceeding the offline approximate optimum.

Table 1: Main performance results on RouterBench ($N = 10,000$) and SPROUT ($N = 13,273$) under budget factor $1.0\times$ allocated by cost-efficiency split. Perf denotes total score, Cost denotes total spending, PPC denotes Perf/Cost, Tput denotes served queries, and RP denotes Relative Performance against approximate offline optimum ($\hat{C}_{opt}$).

Algorithm	RouterBench ($M = 11, N = 10,000$)					SPROUT ($M = 13, N = 13,273$)				
	Perf	Cost	PPC	Tput	RP (%)	Perf	Cost	PPC	Tput	RP (%)
Random	1393.55	0.419	3327.63	3269	43.45	2798.00	0.708	3950.37	4624	46.21
Greedy-Perf	1012.85	0.270	3746.60	1684	31.58	780.90	0.410	1905.82	1096	12.90
Greedy-Cost	1599.75	0.460	3474.51	4033	49.88	3953.00	0.849	4658.56	6752	65.29
KNN-Perf	1005.10	0.271	3715.32	1681	31.34	789.00	0.410	1926.74	1109	13.03
KNN-Cost	1654.55	0.460	3594.42	4049	51.59	3931.50	0.848	4638.31	6647	64.93
BatchSplit	1868.80	0.459	4073.08	3948	58.27	4086.10	0.830	4924.25	6455	67.49
Roberta-Perf	44.75	0.024	1901.21	53	1.40	138.30	0.165	838.35	172	2.28
Roberta-Cost	397.40	0.071	5629.13	972	12.39	831.20	0.115	7224.59	1788	13.73
PORT	2757.75	0.448	6151.78	5213	85.99	4651.30	0.814	5716.69	7720	76.82
RDF-VG (Ours)	3119.30	0.441	7067.32	6093	97.26	7063.35	0.849	8323.21	9739	116.66
Approx Optimum ($\hat{C}_{opt}$)	3207.05	0.460	6972.30	6216	100.00	6054.70	0.850	7125.16	8913	100.00
Optimum (C_{opt})	6376.90	0.460	13865.62	6436	198.84	11934.40	0.849	14060.34	12336	197.11

historical records) and SPROUT Wu & Silwal (2025) (13 candidate LLMs, $N = 13,273$ test queries, $E = 30,968$ historical records). Prompt text is embedded using ‘bge-base-en-v1.5’ Malkov & Yashunin (2016) and indexed with flat cosine HNSW graphs ($|R_j| = 5$).

Budget Allocation & Baselines. The total budget B is set to $1.0\times$ the minimal cost required for a single model to serve all test queries (0.4612 on RouterBench, 0.8490 on SPROUT). The budget is distributed across models proportionally to smoothed historical cost-efficiency (Perf/Cost)^{0.5}. We compare RDF-VG against nine established routing algorithms: Random, Greedy-Perf, Greedy-Cost, KNN-Perf, KNN-Cost Hu et al. (2024), BatchSplit (HIGHS solver, mini-batch size 256) Bumpensanti & Wang (2018), Roberta-Perf, Roberta-Cost Liu et al. (2019), and PORT Wu & Silwal (2025), alongside the offline approximate LP optimum ($\hat{C}_{opt}$) and true integer LP optimum (C_{opt}).

Metrics. We report four primary evaluation metrics: (1) Total Performance (Perf), (2) Total Cost, (3) Performance per Cost (PPC = $\text{Perf} / \text{Cost}$), (4) Throughput (Tput, total queries served), and (5) Relative Performance (RP = $\text{Perf} / \hat{C}_{opt}$).

4.2 MAIN PERFORMANCE RESULTS ON ROUTERBENCH & SPROUT

As shown in Table 1 and Figure 2, RDF-VG achieves superior performance, cost efficiency, and throughput across both benchmarks. On RouterBench, RDF-VG achieves a total performance score

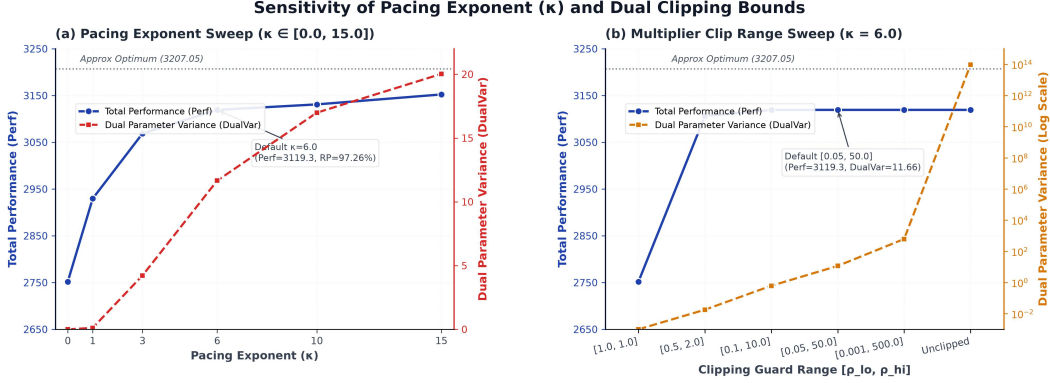


Figure 3: Sensitivity analysis of regularized online dual-pacing parameters on RouterBench. Panel (a) displays Total Performance and Effective Dual Parameter Variance across pacing exponents κ . Panel (b) evaluates clipping guard ranges at $\kappa = 6.0$, demonstrating that clipping guards prevent parameter explosion while maintaining high performance.

Table 2: Sensitivity sweep over pacing exponent κ (top) and dual clipping guard ranges $[p_{lo}, p_{hi}]$ (bottom) on RouterBench. $\kappa = 0.0$ corresponds to static dual weights (γ^*).

Parameter Sweep	Perf	Cost	PPC	Tput	RP (%)	DualVar(g_{eff})
$\kappa = 0.0$ (Static γ^*)	2751.95	0.341	8071.5	4955	85.81	5.58×10^{-27}
$\kappa = 1.0$	2929.50	0.390	7520.5	5603	91.35	1.07×10^{-01}
$\kappa = 3.0$	3068.80	0.427	7194.3	5962	95.69	4.22×10^{00}
$\kappa = 6.0$ (Default)	3119.30	0.441	7067.3	6093	97.26	1.17×10^{01}
$\kappa = 10.0$	3131.05	0.448	6985.9	6138	97.63	1.70×10^{01}
$\kappa = 15.0$	3152.30	0.452	6967.1	6164	98.29	2.00×10^{01}
[1.0, 1.0] (Frozen Multiplier)	2751.95	0.341	8071.5	4955	85.81	5.58×10^{-27}
[0.5, 2.0]	3106.05	0.441	7051.0	6076	96.85	1.81×10^{-02}
[0.05, 50.0] (Default)	3119.30	0.441	7067.3	6093	97.26	1.17×10^{01}
Unclipped	3119.30	0.441	7067.3	6093	97.26	9.64×10^{13}

of 3119.30 (97.26% RP), significantly outperforming PORT (2757.75, 85.99% RP) by +11.27 percentage points RP and BatchSplit (1868.80, 58.27% RP) by +38.99 percentage points RP. RDF-VG also processes 6093 queries compared to PORT’s 5213 queries, approaching the approximate LP oracle (6216 queries).

On SPROUT, RDF-VG reaches 7063.35 total performance (116.66% RP), outperforming PORT (4651.30, 76.82% RP) by +39.84 percentage points RP. The reason RDF-VG exceeds 100% RP relative to the offline approximate LP optimum ($\hat{C}_{opt} = 6054.70$) lies in SPROUT’s high token length variance and capacity bottlenecks. While the static LP oracle allocates expected batch budgets, RDF-VG’s value-gated feasibility fallback dynamically captures unspent capacity from budget-depleted primary models and redirects queries to affordable, high-value alternative models, processing 9739 queries compared to the oracle’s 8913 served queries.

4.3 COMPONENT ABLATIONS & SENSITIVITY ANALYSIS

Ablation 1: Regularized Pacing Exponent & Clipping Bounds. As detailed in Table 2 and Figure 3, online pacing provides a massive +400.35 performance score boost (+12.48 percentage points RP) over static dual control ($\kappa = 0$). Increasing κ smoothly improves performance until saturating around $\kappa = 6.0$. Furthermore, panel (b) shows that unclipped pacing allows dual parameter variance to surge to 9.64×10^{13} , whereas the guard range [0.05, 50.0] stabilizes dual price fluctuations while maintaining maximum routing efficiency.

Table 3: Ablation of fallback policies on RouterBench. Value-gating prevents budget dissipation on low-value model assignments.

Fallback Policy Variant	Perf	Cost	PPC	Tput	RP (%)	Triggers
No Fallback (Standard PORT)	2756.75	0.448	6150.09	5213	85.96	4728
Cost-Greedy Fallback	2056.00	0.461	4462.03	4148	64.11	6916
Performance-Greedy Fallback	2037.35	0.461	4421.43	4114	63.53	6964
Next-Best Dual (Ungated)	2114.15	0.461	4588.06	4155	65.92	6822
Next-Best Dual + Value-Gate (RDF-VG)	2751.95	0.341	8071.48	4955	85.81	217

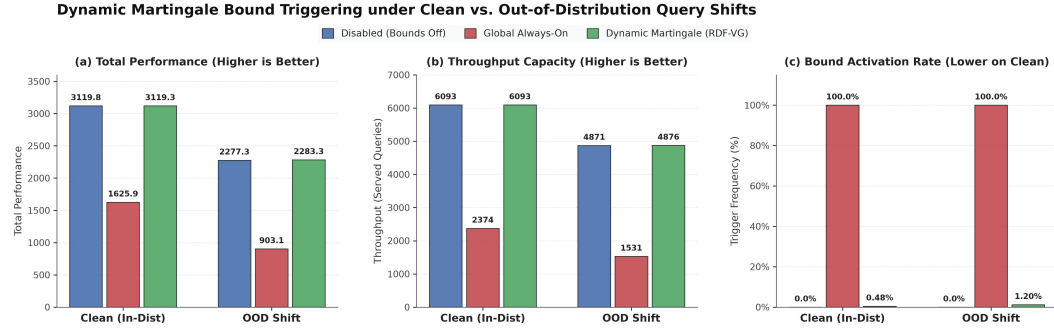


Figure 4: Comparison of martingale bound triggering strategies across clean in-distribution and out-of-distribution (OOD) query shift settings. Dynamic martingale bounds achieve near-zero overhead on clean data (triggering on 0.48% of queries) while selectively activating on 1.20% of queries under distribution shift to preserve optimal performance.

Table 4: Performance of martingale triggering logic across Clean (in-distribution) and OOD (MMLU history $\rightarrow$ non-MMLU test queries) settings on RouterBench.

Martingale Variant	Clean Setting (In-Distribution)					OOD Setting (MMLU $\rightarrow$ Non-MMLU)				
	Perf	Cost	PPC	Tput	Trig (%)	Perf	Cost	PPC	Tput	Trig (%)
1. Disabled (Bounds Off)	3119.80	0.441	7066.91	6093	0.00	2277.30	0.539	4228.11	4871	0.00
2. Global Always-On	1625.85	0.295	5511.60	2374	100.00	903.05	0.281	3217.24	1531	100.00
3. Distance-Only (p90)	3032.95	0.436	6955.41	5852	10.74	1729.80	0.466	3710.55	3273	53.98
4. Variance-Only (p90)	2893.05	0.435	6650.51	5732	18.66	2088.75	0.515	4056.51	4289	30.12
5. Dual Trigger (p90)	2834.05	0.433	6549.91	5523	26.86	1572.55	0.465	3378.63	2999	57.76
6. RDF-VG Dynamic (Default)	3119.30	0.441	7067.32	6093	0.48	2283.30	0.538	4243.36	4876	1.20

Ablation 2: Value-Gated Feasibility Fallback Policies. Table 3 compares five fallback strategies when a model’s budget is exhausted. Raw ungated fallback strategies (Cost-Greedy, Perf-Greedy, Ungated Dual) severely degrade performance (2114.15 Perf) compared to No Fallback (2756.75 Perf) because they deplete remaining model budgets on low-value queries. Introducing value-gating ($\alpha \hat{d}_{ij} - \hat{g}_{ij} g_{\text{eff},i}(t) \geq 0$) reduces primary infeasibility triggers from 6,822 to 217 events, elevating cost efficiency (PPC) from 6150.09 to 8071.48 while preserving high overall performance.

Ablation 3: Dynamic Martingale Trigger Logic under Clean vs. OOD Shift. As shown in Table 4 and Figure 4, applying martingale bounds globally (Global Always-On) severely collapses clean performance from 3119.80 to 1625.85 due to extreme bound conservatism. In contrast, RDF-VG’s dynamic tail triggering activates on only 0.48% of clean queries, matching the unprotected baseline (3119.30 vs 3119.80 Perf). Under severe OOD distribution shift (MMLU historical logs $\rightarrow$ non-MMLU test prompts), RDF-VG selectively triggers on 1.20% of atypical queries, delivering positive gain (+6.00 Perf, 2283.30 vs 2277.30) without suffering throughput collapse.

Ablation 4: Warm-Up Fraction ϵ vs. Online Learning Rate η . Table 5 and Figure 5 illustrate the interplay between the offline warm-up ratio ϵ and online pacing learning rate η . When online pacing is disabled ($\eta = 0$), performance remains capped below 2751.9, whereas small step sizes

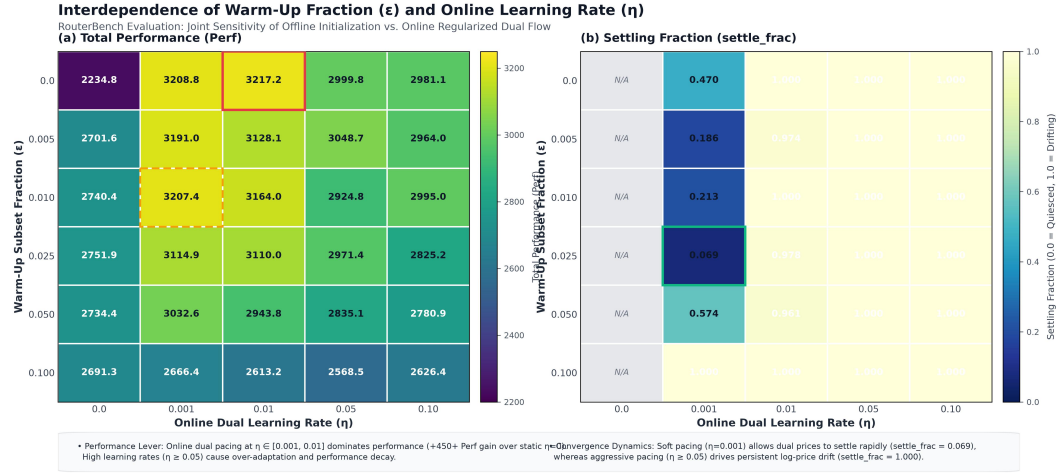


Figure 5: Interdependence of warm-up fraction (ϵ) and online dual learning rate (η) on RouterBench. (a) Total Performance demonstrates that moderate online learning rates ($\eta \in [0.001, 0.01]$) yield optimal routing scores, while excessive rates ($\eta \geq 0.05$) degrade performance. (b) Dual parameter settling fraction shows rapid parameter convergence under soft pacing.

Table 5: Heatmap grid mapping Total Performance (Perf) across warm-up fractions ϵ and online step sizes η on RouterBench.

Warm-Up Fraction (ϵ)	$\eta = 0.0$	$\eta = 0.001$	$\eta = 0.01$	$\eta = 0.05$	$\eta = 0.10$
$\epsilon = 0.005$	2234.8	3208.8	3217.2	2999.8	2981.1
$\epsilon = 0.010$	2701.6	3191.0	3128.1	3048.7	2964.0
$\epsilon = 0.025$ (Default)	2740.4	3207.4	3164.0	2924.8	2995.0
$\epsilon = 0.050$	2751.9	3114.9	3110.0	2971.4	2825.2
$\epsilon = 0.100$	2734.4	3032.6	2943.8	2835.1	2780.9
$\epsilon = 0.100$	2691.3	2666.4	2613.2	2568.5	2626.4
$\epsilon = 0.000$ (Pure Online)	2234.8	3208.8	3217.2	2999.8	2981.1

Table 6: Robustness comparison across budget split schemes ($1.0\times$ budget, top) and total budget scaling factors (cost-efficiency split, bottom) on RouterBench.

Budget Split Strategy	BatchSplit Perf	PORT Perf	RDF-VG Perf	RDF-VG RP (%)	Gain vs PORT
Cost-Efficiency (Default)	1868.80	2757.75	3119.30	97.26	+361.55
Uniform Split	1403.95	2148.35	2720.05	96.44	+571.70
Performance-Based Split	1246.60	1943.90	2633.80	96.09	+689.90
Cost-Based Split	1918.80	2641.85	3089.45	96.54	+447.60
Extreme Split (80 $\rightarrow$ 1 least-eff)	441.40	711.30	1085.65	83.58	+374.35
Extreme Split (80 $\rightarrow$ 3 least-eff)	500.65	727.70	1172.25	81.73	+444.55
Extreme Split (80 $\rightarrow$ 5 least-eff)	822.25	1119.70	1727.20	88.54	+607.50
Total Budget Multiplier	BatchSplit RP	PORT RP	RDF-VG RP	RDF-VG Perf	Gain vs PORT
0.25 $\times$ Total Budget	36.49%	54.92%	86.76%	1163.50	+426.95
0.50 $\times$ Total Budget	41.77%	65.50%	95.20%	2107.55	+657.50
1.00 $\times$ Total Budget	58.27%	85.99%	97.26%	3119.30	+361.55
1.50 $\times$ Total Budget	67.53%	88.89%	96.73%	3886.10	+314.85
2.00 $\times$ Total Budget	77.65%	93.14%	97.09%	4568.45	+186.00

($\eta \in [0.001, 0.01]$) consistently lift performance across all warm-up fractions. Notably, under pure online bootstrapping ($\epsilon = 0.0$), $\eta = 0.01$ achieves 3217.2 Perf, showing that RDF-VG’s closed-loop pacing can learn optimal shadow prices from scratch.

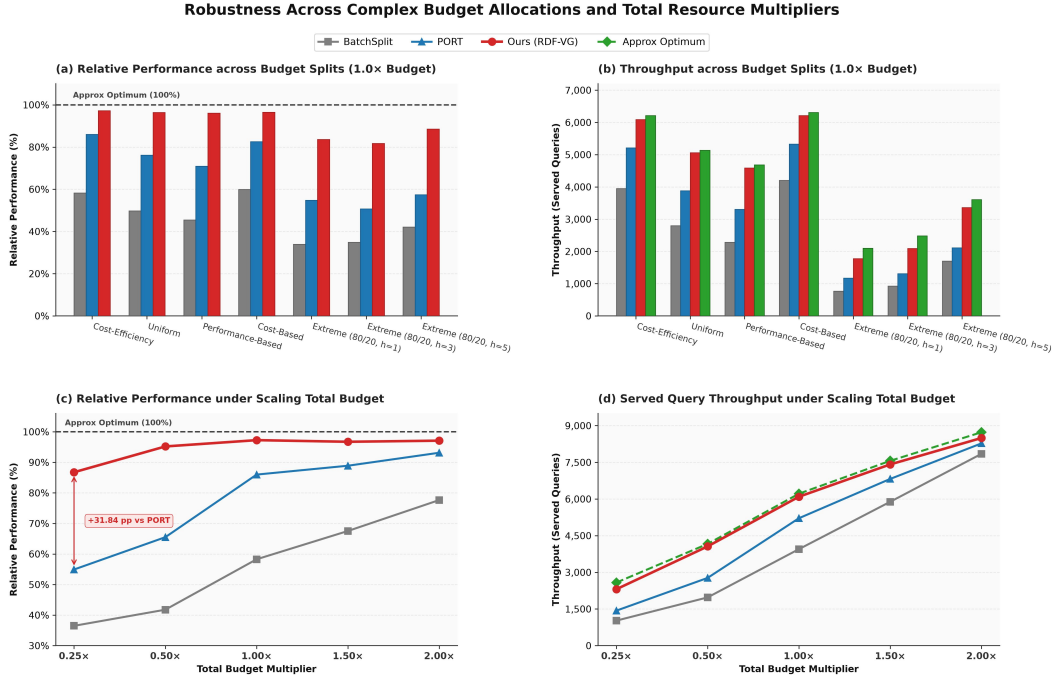


Figure 6: Robustness across budget allocations and total resource multipliers on RouterBench. Panels (a) and (b) compare Relative Performance and throughput across seven budget split schemes at $1.0\times$ budget. Panels (c) and (d) evaluate performance and throughput scaling under budget multipliers from $0.25\times$ to $2.00\times$.

4.4 ROBUSTNESS TO BUDGET CONSTRAINTS & SYSTEM SCALE

Ablation 5: Budget Allocation Schemes and Total Resource Multipliers. As shown in Table 6 and Figure 6, RDF-VG outperforms PORT and BatchSplit across all seven budget split strategies and all five scaling multipliers. Crucially, under extreme scarcity ($0.25\times$ budget), RDF-VG achieves 86.76% RP (1163.50 Perf), whereas PORT collapses to 54.92% RP and BatchSplit to 36.49% RP.

Ablation 6: Historical Data Volume $|D|$ and Candidate Pool Size $|R_j|$. RDF-VG is robust to aggressive repository compression—subsampling records $|D|$ from 26,497 to 5,000 retains 97.44% RP—and sustains sub-0.3 ms end-to-end latency across all pool sizes $|R_j|$ (0.2855 ms total: 0.2236 ms retrieval + 0.0619 ms decision), confirming suitability for high-throughput serving. Full profiling is deferred to Table 7 in Appendix B.

5 CONCLUSION

We introduced **RDF-VG**, a training-free online multi-LLM router for budget-constrained serving that resolves open-loop static routing failures via regularized online dual-pacing and value-gated feasibility fallback, anchored to global LP shadow prices and guarded by dynamically-triggered empirical Bernstein bounds under distribution shift. It attains 97.26% and 116.66% of the approximate offline LP optimum on RouterBench and SPROUT at sub-0.3 ms per-query overhead; multi-objective SLA optimization and federated dual-parameter synchronization are natural next steps.

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Table 7: Impact of historical data volume $|D|$ (top) and candidate pool size $|R_j|$ (bottom) on routing quality and per-query latency components on RouterBench (single-thread wall-clock timings in ms).

Subsampling Sweep	Perf	Cost	PPC	RP (%)	Retrieval (ms)	Routing (ms)	Total Latency
$ D = 2,000$	3050.15	0.445	6848.70	95.11	0.1002	0.0635	0.1637 ms
$ D = 5,000$	3125.00	0.447	6987.68	97.44	0.1296	0.0650	0.1947 ms
$ D = 10,000$	3111.90	0.443	7018.98	97.03	0.1566	0.0627	0.2194 ms
$ D = 20,000$	3149.70	0.446	7062.09	98.21	0.2074	0.0635	0.2709 ms
$ D = 26,497$ (Full)	3119.30	0.441	7067.32	97.26	0.2226	0.0646	0.2871 ms
$ R_j = 1$	2848.70	0.445	6405.28	88.83	0.2205	0.0759	0.2964 ms
$ R_j = 3$	3123.90	0.441	7086.60	97.41	0.2231	0.0621	0.2851 ms
$ R_j = 5$ (Default)	3119.30	0.441	7067.32	97.26	0.2236	0.0619	0.2855 ms
$ R_j = 10$	3202.70	0.442	7248.73	99.86	0.2265	0.0631	0.2896 ms
$ R_j = 20$	3222.60	0.441	7308.34	100.48	0.2301	0.0675	0.2976 ms

A THEORETICAL DERIVATIONS & MARTINGALE BOUNDS

In this section, we provide formal theoretical derivations for LP dual relaxation convergence and time-uniform empirical Bernstein supermartingale confidence sequences.

A.1 LP DUAL CONVERGENCE AND PRIMAL-DUAL BOUND

Let C_{opt} denote the optimal objective value of the integer linear program (Equation 1) over query stream $\mathcal{Q}$, and let $\hat{C}_{\text{opt}}$ denote the optimal objective value of the relaxed linear program (Equation 4). Following standard online linear programming theory Li & Ye (2019), under independent and identically distributed query arrivals drawn from unknown distribution $\mathcal{D}_q$, the empirical dual shadow price vector γ^* learned over warm-up subset $\mathcal{P}$ ($|\mathcal{P}| = \epsilon N$) satisfies:

$$\mathbb{E} \left[\frac{1}{N} \sum_{j=1}^N \sum_{i=1}^M \alpha d_{ij} x_{ij}^* \right] \geq (1 - O(\sqrt{\epsilon})) \frac{\hat{C}_{\text{opt}}}{N} - O\left(\frac{S_{\max}}{\epsilon N}\right), \quad (10)$$

where $S_{\max} = \max_j \max_i d_{ij}$ bounds maximum single-query utility score.

A.2 TIME-UNIFORM EMPIRICAL BERNSTEIN SUPERMARTINGALES

To establish any-time valid confidence bounds uniform over unbounded time horizons $t \in \{1, 2, \dots, \infty\}$, we construct empirical Bernstein supermartingales Howard et al. (2018). Let $\{Y_\tau\}_{\tau=1}^t$ be bounded random variables with mean μ and sample variance $\sigma_t^2 = \frac{1}{t} \sum_{\tau=1}^t (Y_\tau - \hat{\mu}_t)^2$. For any target error probability $\delta \in (0, 1)$, the time-uniform boundary $\mathcal{V}_t(\delta)$ satisfies:

$$\mathbb{P} \left(\exists t \geq 1 : |\hat{\mu}_t - \mu| \geq \sqrt{\frac{2\sigma_t^2 \log\left(\frac{\zeta(1.6) \log^{1.6}(t)}{\delta}\right)}{t}} + \frac{3M_y \log\left(\frac{\zeta(1.6) \log^{1.6}(t)}{\delta}\right)}{t} \right) \leq \delta. \quad (11)$$

Applying this inequality to candidate neighbor sets R_j yields the non-asymptotic bounds $g_{ij}^{\text{upper}}(t)$ and $d_{ij}^{\text{lower}}(t)$ defined in Section 3.

B EXTENDED EXPERIMENTAL LOGS & PER-MODEL COST PROFILING

We first report the full per-configuration sweep over historical data volume $|D|$ and candidate pool size $|R_j|$ referenced in Section 4 (Ablation 6). Table 8 and Table 9 then present granular per-model historical statistics and pricing structures across candidate LLM pools on RouterBench and SPROUT benchmarks.

Table 8: RouterBench candidate LLM pool statistics computed over historical dataset ($E = 26,497$).

Model Name	Avg Cost (\$)	Avg Perf	Perf / Cost	Smoothed Weight
WizardLM-13B-V1.2	7.27×10^{-5}	0.432	5944	77.10
claude-instant-v1	2.32×10^{-4}	0.598	2581	50.80
claude-v1	2.14×10^{-3}	0.631	295	17.18
claude-v2	2.41×10^{-3}	0.636	264	16.24
gpt-3.5-turbo-1106	2.42×10^{-4}	0.617	2546	50.45
gpt-4-1106-preview	3.28×10^{-3}	0.781	238	15.43
code-llama-instruct-34b	1.71×10^{-4}	0.203	1182	34.38
llama-2-70b-chat	2.02×10^{-4}	0.328	1627	40.34
mistral-7b-chat	4.56×10^{-5}	0.308	6768	82.27
mixtral-8x7b-chat	1.34×10^{-4}	0.550	4098	64.01
Yi-34B-Chat	1.85×10^{-4}	0.648	3503	59.19

Table 9: SPROUT candidate LLM pool pricing and historical statistics ($E = 30,968$).

Model Name	Input Cost (\$/1M)	Output Cost (\$/1M)	Avg Perf	Perf / Cost
claude-3-5-sonnet-v1	3.00	15.00	0.827	108
titan-text-premier-v1	0.50	1.50	0.579	1027
openai-gpt-4o	2.50	10.00	0.846	172
openai-gpt-4o-mini	0.15	0.60	0.808	2378
granite-3-2b-instruct	0.10	0.10	0.553	6473
granite-3-8b-instruct	0.20	0.20	0.659	4403
llama-3-1-70b-instruct	0.90	0.90	0.810	1130
llama-3-1-8b-instruct	0.20	0.20	0.690	2838
llama-3-2-1b-instruct	0.06	0.06	0.460	6904
llama-3-2-3b-instruct	0.06	0.06	0.629	9722
llama-3-3-70b-instruct	0.90	0.90	0.804	1457
llama-3-405b-instruct	3.50	3.50	0.776	385
mixtral-8x7b-instruct	0.60	0.60	0.616	1648